

UME → GME AI Competency & Integration Map

Detailed Reference — For Dean-Level Review and Faculty Distribution

This document describes a three-phase curriculum framework for integrating AI competency into undergraduate and graduate medical education. It runs from MS1 through PGY2. Each phase is sequenced: students demonstrate AI-independent clinical reasoning before they work alongside AI tools, then build error-detection capacity in supervised adversarial settings, then use AI in live clinical care with hard documentation and escalation requirements.

The framework addresses a specific risk — which the field now calls **never-skilling** — distinct from deskilling in experienced clinicians. Never-skilling is the failure to develop foundational reasoning competencies during formative training when AI substitutes the cognitive effort those competencies require. The concern is grounded in desirable difficulties theory and cognitive load research, with early empirical signals from nonclinical settings. Direct causal evidence from clinical trainees is not yet available. This framework is precautionary, not evidence-established.

Every milestone in this document names a clinical context, a measurable observable behavior, a required artifact, an assessment method, an oversight level, risk controls, and explicit LCME/ACGME alignment.

Primary sources

Ke et al. *Nature Medicine* 2026 (never-skilling framework) · Hunt et al. *npj Digital Medicine* 2026 (54-study UME competency taxonomy) · Abdulnour, Gin, Boscardin. *NEJM* 2025 (DEFT-AI supervisory framework) · AAMC Principles for Responsible AI Use v2.0, July 2025 · ACGME CPR active stakeholder consultation, AI named in 8 topic areas, formal guidance expected 2026–27

PHASE 1 · MS1 – MS2

AI-Independent Baseline

Students build clinical reasoning before they work alongside AI. Every assessed case runs in a lockdown environment — no LLMs, no CDS alerts, no predictive templates. AI is allowed as a study tool or Socratic prompt generator. It is prohibited wherever it supplies the answer the student is supposed to construct.

Why it comes first: Schema formation requires effortful processing in working memory. When AI delivers correct answers before a student has built the reasoning structure, the cognitive work that produces durable learning does not occur. This is not a blanket ban — it is a protected developmental window.

■ PROGRESSION GATE

AI-restricted OSCE ≥75% · Calibrated self-assessment within 1 point of faculty rating

Accreditation alignment: LCME 6.3 · 7.1 · 7.6 · ACGME MK · PBLI-1 · Prof

1. Model Fundamentals — Offline Didactic

Clinical Context	Informatics module (MS1, year 1, semester 1)
Milestone	Complete a 90-min interactive module. Score ≥80% on embedded 10-question quiz. Produce a 1-page concept map linking tokenization, embeddings, context window, hallucination, and temperature each to a distinct clinical failure mode example.
Required Artifact	Quiz score (LMS record) + concept map (faculty-reviewed)
Assessment Method	Proctored online module quiz + faculty concept map review
Faculty Oversight	Direct
Risk Controls	Module uses offline AI simulator only — no live LLM, no PHI. Faculty attestation required.

2. AI-Independent Differential Diagnosis

Clinical Context	Preclinical case-based learning lab (MS1–MS2, proctored)
Milestone	Given a written clinical vignette, generate a ranked differential of ≥5 conditions without any AI tool access (lockdown browser or paper). ≥4/5 differentials clinically defensible per faculty key. Verbally defend each entry in structured debrief.
Required Artifact	Handwritten or paper differential + signed zero-AI-use attestation
Assessment Method	Proctored written OSCE (AI-restricted) + oral debrief using DEFT-AI structure
Faculty Oversight	Direct
Risk Controls	Lockdown browser enforced. Phone use during exam = academic integrity violation (policy must be explicit). No demographic anchoring cues in vignette.

3. Privacy, HIPAA, and AI Data Risk

Clinical Context	Preclinical ethics seminar (MS2)
Milestone	Given a scenario where a classmate submits PHI into a public commercial LLM, identify the HIPAA violation, the institutional escalation trigger, and a de-identification protocol that would permit safe use in an approved institutional LLM instance. Faculty rubric ≥3/4.
Required Artifact	1-page written scenario analysis + signed AI responsible-use policy acknowledgment
Assessment Method	Proctored written analysis or proctored viva
Faculty Oversight	Direct
Risk Controls	Scenario uses fabricated vignette — no real patient data. Faculty models correct de-identification live. Equity lens: discuss how stripping demographics for de-identification may eliminate data needed for equitable care.

4. AI-Independent Standardized Patient Encounter

Clinical Context	Clinical reasoning lab — standardized patient simulation (MS2)
Milestone	20-min history and physical, AI-restricted. Problem list + management plan for top problem. Self-assessed confidence within 1 point of faculty-assigned performance rating. SP checklist ≥75%.
Required Artifact	SP checklist + problem list + management plan + student self-assessment confidence form + discrepancy report if gap ≥2 points
Assessment Method	Observed SP OSCE (AI-restricted); faculty calibration debrief
Faculty Oversight	Direct
Risk Controls	SP demographics standardized across race, sex, age, body habitus to prevent anchoring. Faculty observe for covert device use. AI-independent baseline logged for longitudinal comparison.

PHASE 2 · MS3 – MS4

Adversarial Calibration

Students encounter AI in clinical environments, but always with a documented reasoning trace. The Dual-Reasoning Reflection Log requires an independent written assessment before the AI output appears. Adversarial workshops train error detection across three failure-mode categories before high-stakes practice.

Why adversarial training matters: AI errors in real clinical settings are unlabeled, plausible, and embedded in otherwise coherent reasoning. Students who have only seen AI succeed will accept errors they cannot detect. Structured adversarial exposure — before these errors can influence patient care — builds the calibration infrastructure that safe AI use requires.

■ PROGRESSION GATE	≥8/9 embedded errors identified in adversarial workshop · DRRL completed for 100% of assessed clinical cases
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Accreditation alignment: LCME 7.4 · 7.6 · 7.9 · ACGME PC-1 · MK · Prof-3 · PBLI-3 · SBP-4

1. Inpatient Admit — Dual-Reasoning Reflection Log

Clinical Context	Inpatient medicine clerkship — morning admit, teaching ward (MS3)
Milestone	Before any AI output is displayed, document independent differential and plan (handwritten, timestamped). Then review EHR AI recommendation. Complete DRRL: ≥2 items accepted with evidence citation, ≥1 item modified or overridden with reasoning. Final admitting note distinguishes AI-assisted from student-generated content. DRRL must be filed within 60 minutes of note submission.
Required Artifact	Handwritten pre-AI draft (timestamped) + completed DRRL + attending-countersigned note with AI disclosure section
Assessment Method	Direct observation of note-writing process; attending chart audit against DRRL; end-of-rotation DRRL portfolio review

Faculty Oversight	Direct
Risk Controls	Institutional LLM instance only (no PHI to public cloud). Note cannot be submitted without DRRL. Zero-override note triggers immediate debrief. Faculty review AI differentials for demographic bias (e.g., pain underassessment in Black patients).

2. Outpatient RAG Pipeline Audit

Clinical Context	Primary care clerkship — chronic disease follow-up (MS3)
Milestone	Before viewing AI risk stratification output, document independent risk assessment and plan (≤ 15 minutes). Audit ≥ 2 RAG source citations attached to the AI recommendation — verify each is retrievable and current. If AI tool incorporates race-based correction factors, document whether use is appropriate for this patient with explicit reasoning. Preceptor co-signs visit note same session.
Required Artifact	Pre-visit independent assessment note + RAG audit checklist (≥ 2 sources verified) + race-based algorithm analysis + attending co-signed note
Assessment Method	Adversarial chart audit comparing pre-AI and post-AI plans; RAG checklist review; end-of-clerkship portfolio
Faculty Oversight	Direct
Risk Controls	RAG audit required before action — no direct copy-paste without source verification. Race-based algorithm flag triggers preceptor review before order. Equity guardrail: flag when AI recommendation diverges from SDOH context or patient-stated preference.

3. Adversarial AI Debugging Workshop

Clinical Context	Late clinical clerkship — structured adversarial session, simulation center (MS3–MS4)
Milestone	3 cases $\times$ 3 embedded error types (factual hallucination, demographic/bias error, RAG source mismatch) — all faculty-validated before use. Identify $\geq 8/9$ error-type combinations. For each, produce a written refutation citing a peer-reviewed source or institutional guideline. Submit Workshop Error Report within 30 minutes of session end.
Required Artifact	Workshop Error Report (3 annotated AI notes + refutations with source citations) + faculty rubric score
Assessment Method	Proctored adversarial chart audit; faculty scoring; group debrief with error pattern review
Faculty Oversight	Direct
Risk Controls	Embedded errors pre-validated by faculty clinician AND informaticist — no unvalidated AI content used as teaching material. Demographic bias case is mandatory in error set. Debrief explicitly notes real-world AI errors are unlabeled and intermixed with plausible reasoning.

4. Discharge Summary with AI Scribe — Safety Audit

Clinical Context	Inpatient discharge planning (MS3–MS4, supervised clerkship)
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Milestone	Use institutional GenAI scribe to draft discharge summary. Review draft and flag ≥ 1 hallucinated or incorrect element using a Hallucination Incident Report form. Correct all flagged elements with EHR source citations. Append a prompt disclosure log (model name, exact prompt, verification steps). Zero uncorrected AI errors in final note submitted to patient chart.
Required Artifact	AI draft (unedited, preserved) + Hallucination Incident Report (if applicable) + corrected final note + prompt disclosure log + attending countersignature
Assessment Method	Attending chart audit comparing AI draft to final note; incident report review; end-of-clerkship aggregate safety report
Faculty Oversight	Direct
Risk Controls	Institutional LLM only. No draft in chart without attending review. Incident reports aggregate for AI safety surveillance. Equity check: final note reviewed for omission of SDOH-relevant follow-up (pharmacy access, transportation, language).

PHASE 3 · PGY1 – PGY2

Supervised Integration

Residents use AI tools in live clinical care under attending supervision. Every AI recommendation requires a documented human reasoning trace before it can influence an order. Hard escalation stops — 15 minutes for safety-relevant divergence — prevent passive relay of AI outputs. Governance participation builds the institutional fluency physician leaders will need within five years.

Why governance participation matters: Residents who use AI tools without understanding how they are evaluated and governed develop a dependency they cannot critically examine. Governance participation gives them the vocabulary — responsible use checklists, risk analysis reports, demographic disparity audits — that will define the physician leadership role in AI-enabled health systems.

PROGRESSION GATE	AI Disclosure Statement in $\geq 95\%$ of notes · ≥ 1 documented override/week · ≥ 2 governance meetings/year
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Accreditation alignment: LCME 7.4 · 7.9 · ACGME SBP-1,4 · PC-1,2,3 · PBLI-2,4 · Prof-1,3

1. Daily Rounds — AI Disclosure Statement

Clinical Context	Inpatient internal medicine, teaching service (PGY1 — direct supervision)
Milestone	Verbalize independent assessment and plan before AI recommendation appears to the team. Document AI Disclosure Statement (≤ 2 sentences: tool used, output accepted/modified/overridden, rationale) in $\geq 95\%$ of daily progress notes during any 2-week audit window. ≥ 1 meaningful override per week with attending-validated reasoning. Zero AI-recommendation orders enacted without attending review.
Required Artifact	Daily progress notes (2-week audit sample) + weekly override log with attending validation + escalation records

Assessment Method	Direct observation during rounds; attending chart audit; weekly override log review; ACGME Milestone rating end of rotation
Faculty Oversight	Direct
Risk Controls	Attending countersigns any order modified by AI recommendation. Zero-override across ≥10 cases triggers escalation debrief (proxy for over-reliance). Program director reviews override patterns by patient demographics quarterly.

2. ED Sepsis Alert — Escalation Protocol

Clinical Context	Emergency medicine clerkship or rotation (MS4 / PGY1, direct supervision)
Milestone	On receiving a CDS sepsis alert, document an independent Sepsis-3 assessment within 15 minutes (HR, RR, temp, WBC, lactate, mental status, BP). Estimate personal probability (low/medium/high) and intended action. If student/resident assessment conflicts with alert severity, notify attending within 10 minutes and document conflict explicitly. No patient management action without attending approval.
Required Artifact	Timestamped pre-alert independent assessment note + conflict documentation log + escalation record + attending-signed plan
Assessment Method	Direct observation by ED attending; chart audit of timestamped entries; 30-day alert response audit
Faculty Oversight	Direct
Risk Controls	10-minute escalation is a hard stop — not a guideline. Alert-level conflict not escalated within 10 min triggers safety review. Sepsis model false-positive rate for patients with chronic baseline abnormalities (COPD, CHF) explicitly discussed at rotation orientation. Equity alert: residents briefed that sepsis model may under-alert for patients with atypical presentations.

3. Medication Reconciliation — Safety Log

Clinical Context	Hospital medicine admissions (PGY1, ≥5 medications, direct/indirect supervision)
Milestone	Obtain independent medication history from patient, caregiver, or pharmacy before viewing EHR AI-reconciled list. Document ≥1 discrepancy or explicit concordance statement in Medication Safety Reconciliation Log. If AI list contains high-risk drug (anticoagulant, insulin, narrow therapeutic index), escalate to attending pharmacist co-management within 30 minutes. 100% of admits with ≥5 medications during any 30-day audit window.
Required Artifact	Independent medication history (timestamped) + Medication Safety Reconciliation Log + high-risk escalation records + 30-day safety report
Assessment Method	Direct observation for high-risk cases; MSRL audit (30-day sample); monthly patient safety report to program director
Faculty Oversight	Direct (high-risk) / Indirect (standard)

Risk Controls	High-risk escalation hard stop: anticoagulant or narrow-TI drug not co-signed by pharmacist within 30 minutes triggers attending page. Zero-discrepancy MSRL without concordance statement flagged for review. Language-concordant medication history required when AI may not capture non-English medication names.
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4. AI Governance Committee Participation

Clinical Context	Residency program — institutional AI governance (PGY1–PGY2, protected academic time)
Milestone	Attend ≥2 institutional AI Governance Committee meetings per academic year. Complete ≥1 AI Responsible Use Checklist for a tool used on their service. Present one 5-minute QI case report at a program conference: one real AI tool error, patient impact, corrective action, proposed system-level mitigation. Faculty scoring ≥3/4 on structured rubric.
Required Artifact	Committee attendance records + completed Responsible Use Checklist + QI presentation slides + faculty scoring rubric + AI Safety Incident Report (if applicable)
Assessment Method	QI conference presentation with faculty panel; committee attendance verified by program coordinator; end-of-year portfolio review
Faculty Oversight	Governance Board
Risk Controls	Governance committee includes informaticist, patient safety officer, legal/compliance, patient/community representative. QI presentations de-identified before presentation. Equity mandate: committee charter requires demographic disparity review of AI tool performance twice annually.

Pre-Circulation Checklist

- LCME liaison must validate specific element mapping (6.3, 7.1, 7.4, 7.6, 7.9) against your current LCME self-study before this document is cited in accreditation materials.
- Informatics team must confirm whether your EHR system can enforce a timestamped 'document before AI display' workflow. If not, a paper-based parallel log is the documented workaround.
- Sex-based ACS / pain assessment bias examples on equity slide — confirm primary sources before presenting to faculty.
- Obtain institution-specific data on demographic disparity in EHR AI tool performance. National literature is a placeholder; local data will be more persuasive to clinical faculty.
- Faculty development plan for adversarial case design and validation is a prerequisite for Phase 2 — not an implementation detail. Approximately 9% of medical faculty report AI expertise (Ke et al. 2026).